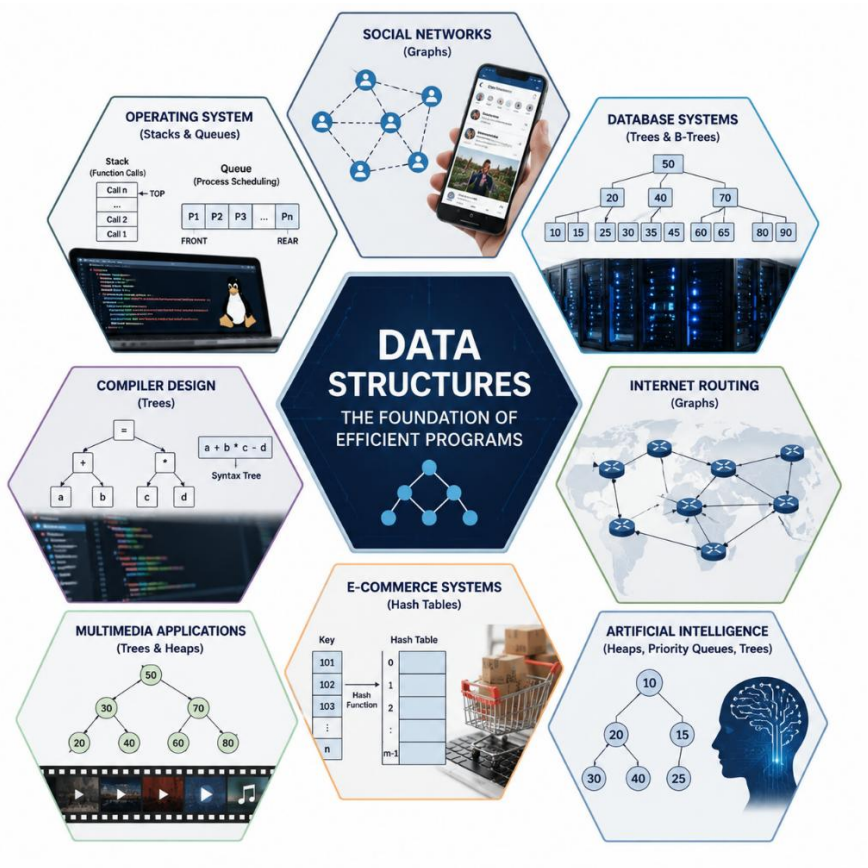
Data Structures & Algorithms Lab

ECE2132



Department of Electronics & Communication Engineering

By:

Dr. D.V. Prashant
Dr. Kamal Kishore Upadhyay

Manipal University Jaipur,

Dehmi Kalan, Off Jaipur-Ajmer Expressway, Jaipur, (Raj.)

Rajasthan 303007

MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

(AY 2026-27)

EXPERIMENT NO. 1

WRITE A PROGRAM IN C++ TO IMPLEMENT POINTERS.

AIM: To understand the concept of pointers in C++ by storing the address of a variable in a pointer and accessing the variable through the pointer.

SOFTWARE USED: Visual Studio Code / Code: Blocks / Dev-C++, GCC/G++Compiler

THEORY: A pointer is a variable that stores the memory address of another variable. Instead of storing data directly, a pointer stores the location where the data is kept in memory. Pointers are widely used in data structures such as linked lists, trees, graphs, stacks, queues, and for dynamic memory allocation. The address-of operator (&) returns the memory address of a variable, while the dereference operator (*) accesses the value stored at that address.

ALOGRITHM:

1. Start the program.
2. Declare an integer variable.
3. Declare a pointer variable.
4. Store the address of the integer variable in the pointer.
5. Display the value and address of the variable.
6. Display the value using the pointer.
7. Stop the program.

PROGRAM (C++):

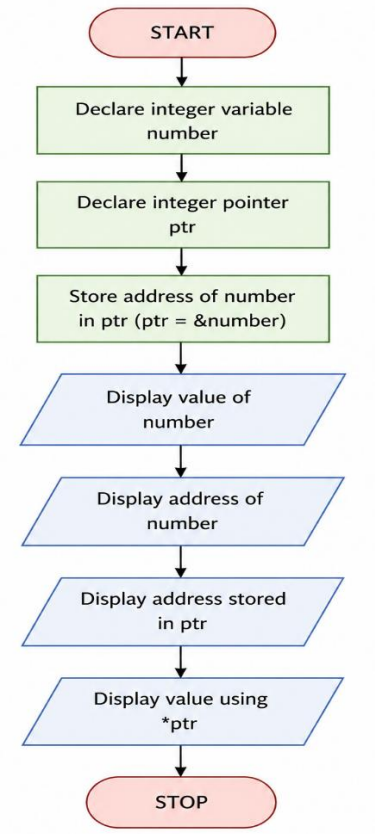
```
#include <iostream>
using namespace std;

int main()
{
    int number = 25;
    int *ptr;

    ptr = &number;

    cout << "Value of number = " << number << endl;
    cout << "Address of number = " << &number << endl;
    cout << "Pointer stores = " << ptr << endl;
    cout << "Value using pointer = " << *ptr << endl;

    return 0;
}
```

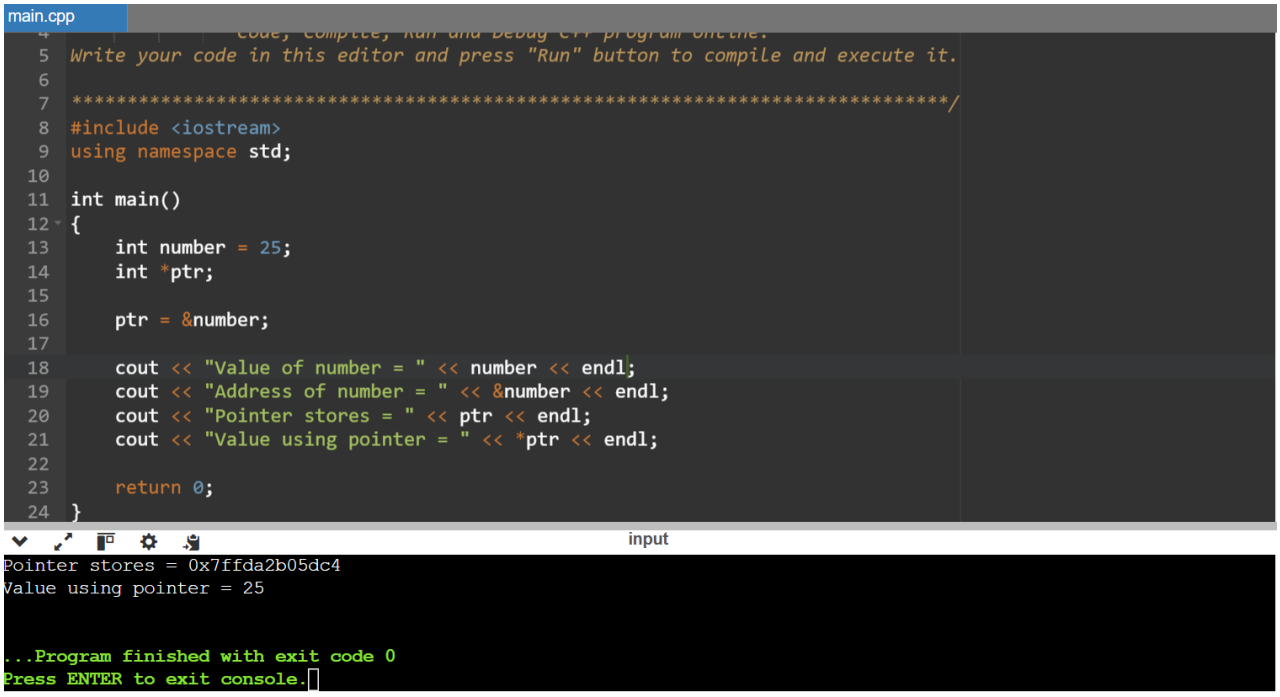


EXPLANATION OF THE PROGRAM:

Statement	Explanation
#include <iostream>	Provides input/output functionality.
using namespace std;	Allows use of standard library names without std::.
int number = 25;	Declares and initializes an integer variable.
int *ptr;	Declares an integer pointer.
ptr = &number;	Stores the address of number in ptr.
&	Address-of operator.
*	Dereference operator used to access the value.

Sample Input: No user input is required.
Sample Output: Value of number = 25, Address of number = 0x61FF08, Pointer stores = 0x61FF08
Value using pointer = 25 (Note: Memory address varies on each system.)
Time Complexity: O(1)
Space Complexity: O(1)

IMPLEMENTATION OF CODE:



APPLICATIONS: Dynamic memory allocation, Linked Lists, Trees, Graphs, Passing arguments by reference.

RESULT: The program was successfully executed. It demonstrated how a pointer stores the address of a variable and accesses the value through dereferencing.